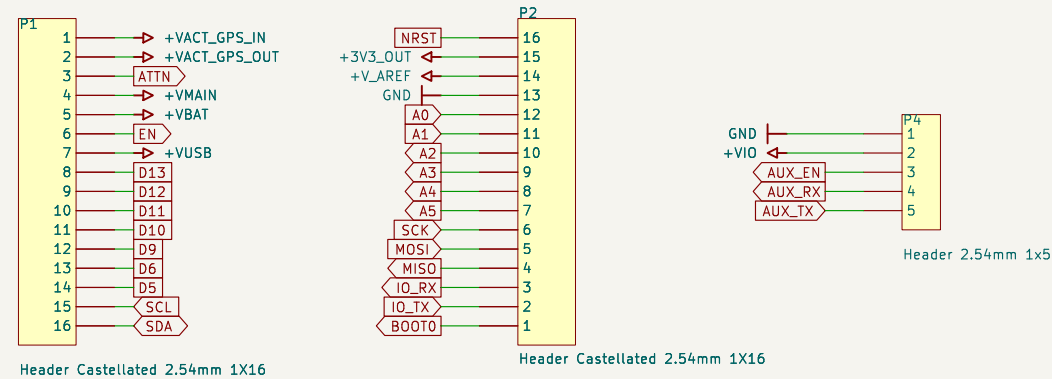


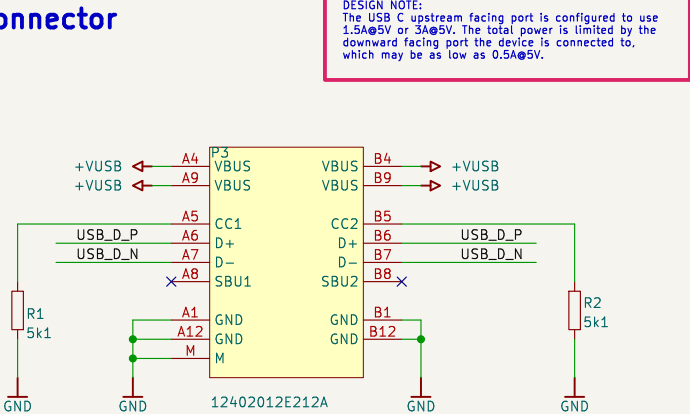
	1	2	3	4	5	6
A	IO Connection File: Notecarrier-CX_IO-Connection.kicad_sch					A
B	Notecard Connection File: Notecarrier-CX_Notecard-Connection.kicad_sch					B
C	Power File: Notecarrier-CX_Power.kicad_sch					C
D	MCU File: Notecarrier-CX_MCU.kicad_sch					D
	1	2	3	4	5	6

Blues Inc	Title: Notecarrier-CX	
	Drawn By: Ported from Altium Designer	
	File: Notecarrier-CX.kicad_sch	
	Rev: v1.7	
Copyright (c) 2023 Blues Inc	Date: 2026-07-28	Sheet: 1/5

I/O Pin Headers



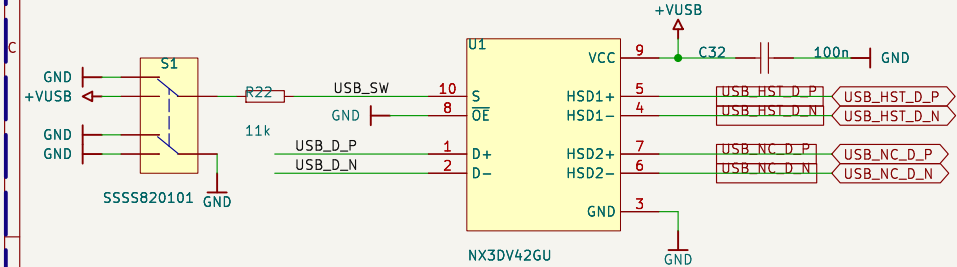
USB C Connector



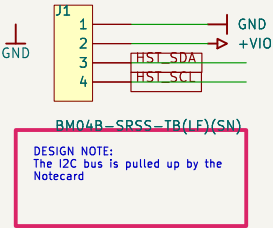
DESIGN NOTE:
The USB C upstream facing port is configured to use 1.5A@5V or 3A@5V. The total power is limited by the downward facing port the device is connected to, which may be as low as 0.5A@5V.

USB Switch

DESIGN NOTE:
The USB connection is routed depending on the position of SW1.
NC (with marker): Host MCU
NO (w/o marker): Notecard

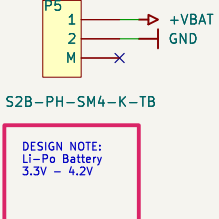


QWIIIC Connector



DESIGN NOTE:
The I2C bus is pulled up by the Notecard

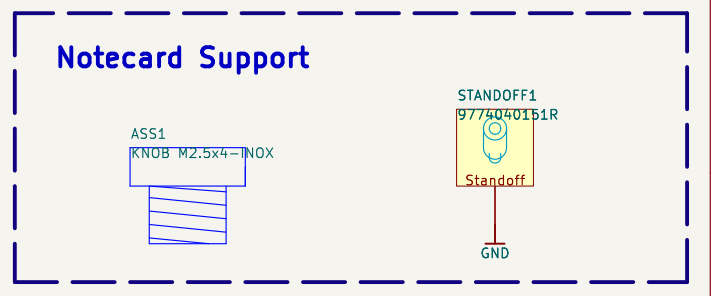
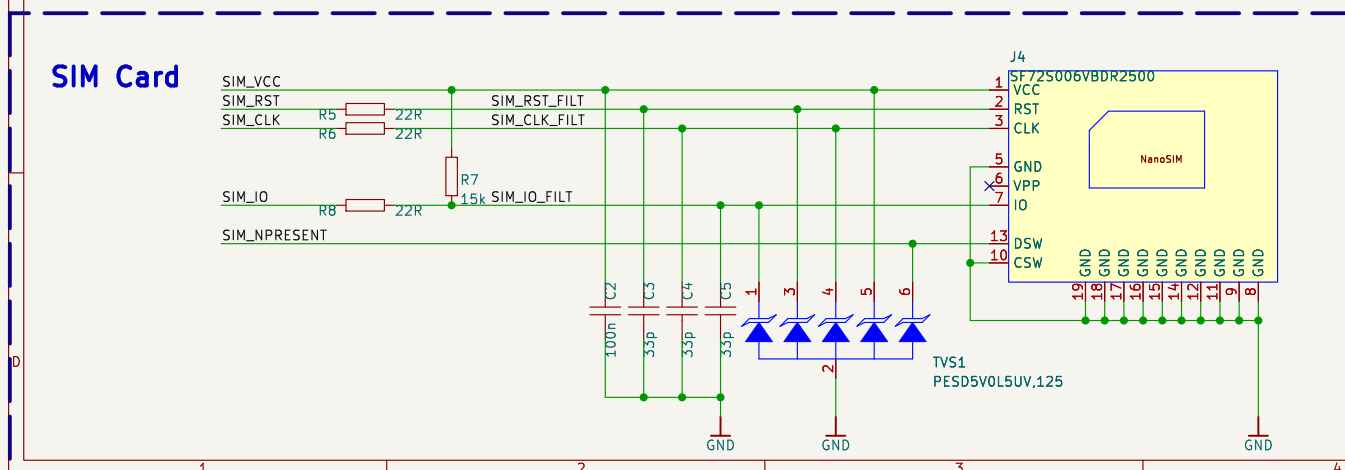
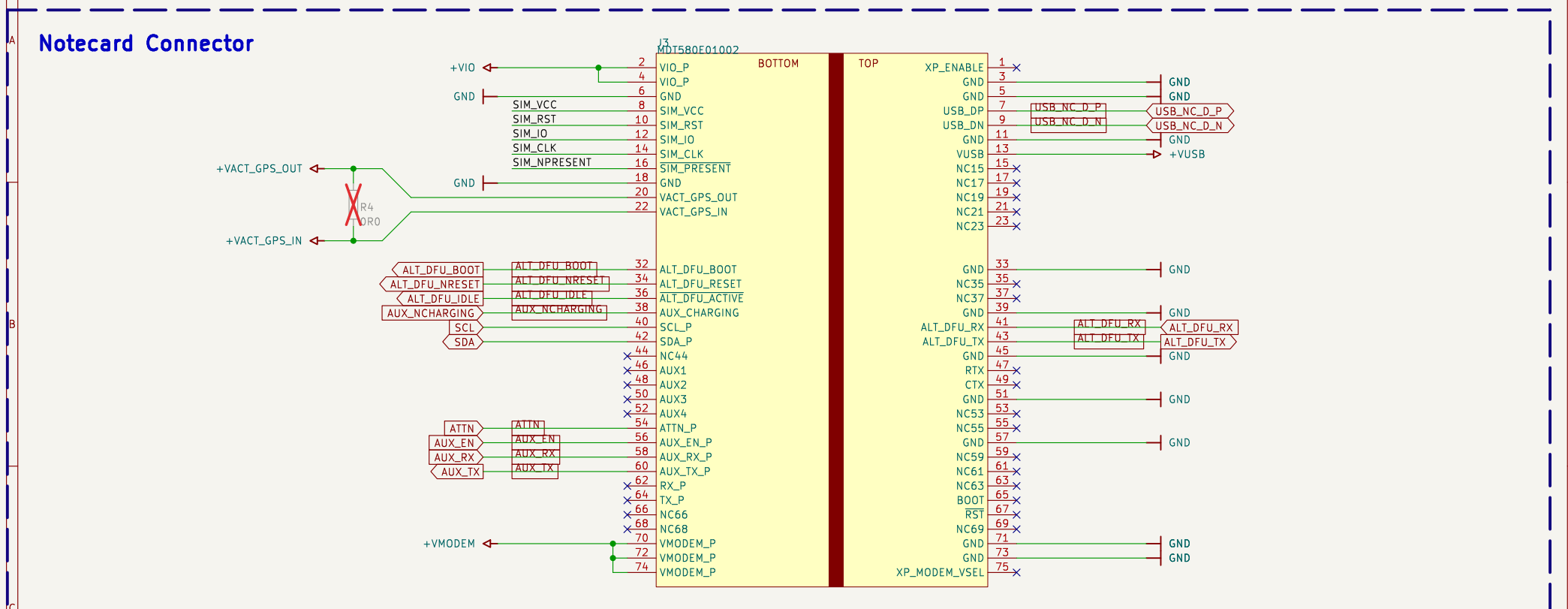
Battery Connector



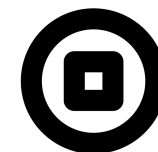
DESIGN NOTE:
Li-Po Battery
3.3V - 4.2V

Title: Notecarrier-CX.kicad_sch	Title:
Project: Notecarrier-CX	Drawn By:
Company: Byte Lab Grupa d.o.o.	Medarska 69/1, 10000 Zagreb, Croatia
File: Notecarrier-CX	www.byte-lab.com
Author: A. Duracki	Rev: = HW_E.Doc.Rev
Copyright (c) 2023 Blues Inc.	Accepted: E. Stanisha
Sheet 2 of 5	Rev: = HW_E.Doc.Rev





Title:	Notecarrier-CX.kicad_sch	Title:	
Project:	Notecarrier-CX	Drawn By:	
Company:	Medarska 69/1, 10000 Zagreb, Croatia	File:	Notecarrier-CX
Byte Lab Grupa d.o.o.	www.byte-lab.com	Copyright (c) 2023 Blues Inc.	Rev: HW-00000000
Author:	A. Duracki	Accepted:	E. Stanisha



Battery Charger w/ Power Path

DESIGN NOTE:

Charger:
 - Default charging voltage: 4.2V
 - Default charging current: 320mA
 - Default termination current: 20mA
 - Default pre-charge current: 20mA
 - No battery thermal protection
 - No battery behaviour: +VBAT is pulled to VUSB - 0.7V through D4 & R20

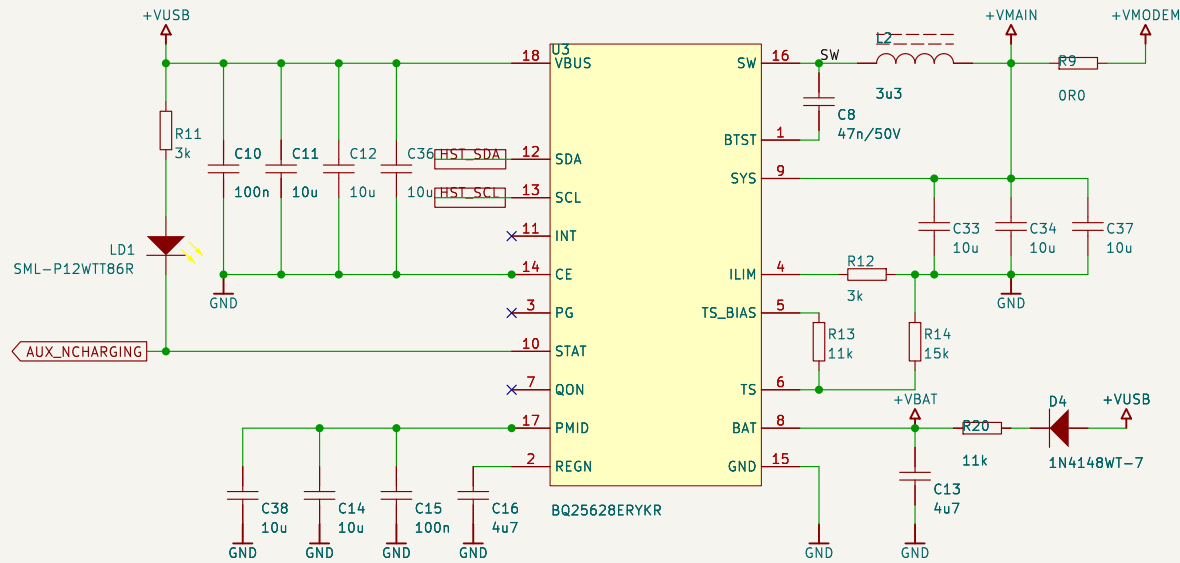
DESIGN NOTE:

Power Path:
 - Input current limit: 3.5A
 - 26mOhm reverse blocking transistor
 - 15mOhm battery transistor

Buck Regulator:
 - 1.5MHz frequency
 - Input Voltage: 4.5-5.5V
 - Output Voltage: 3.3-4.2V
 - Peak current: 1.6A

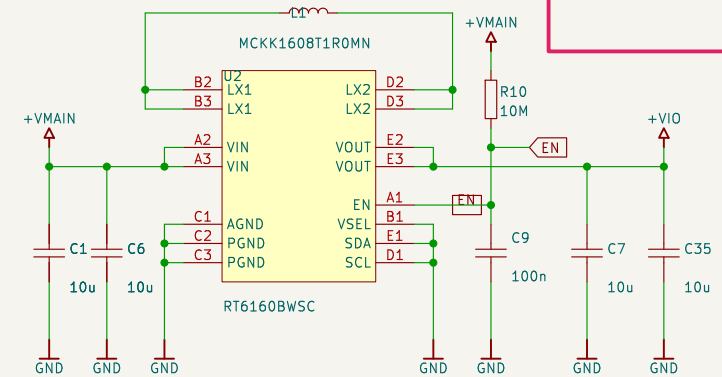
SOFTWARE CONFIG NOTE:

Charge current can be adjusted via I2C registers. Make sure not to exceed 1.5A if programming the charge IC.

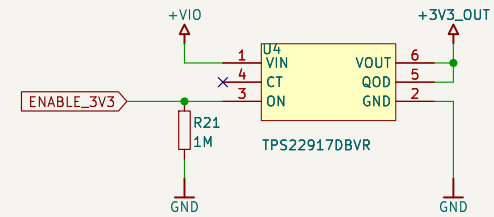


3V3 Buck/Boost Regulator

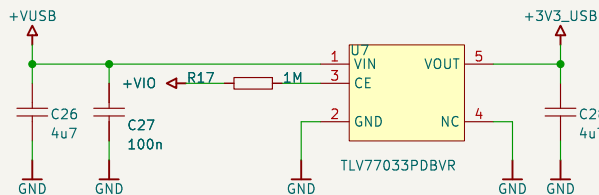
DESIGN NOTE:
 Input Voltage: 2.5V - 5.5V
 Output Voltage: 3.3V
 Max continuous output current over whole input voltage range: 1A



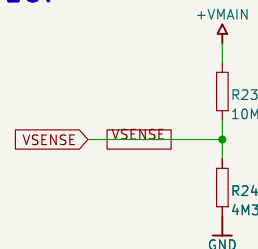
3V3_OUT Switch



USB LDO



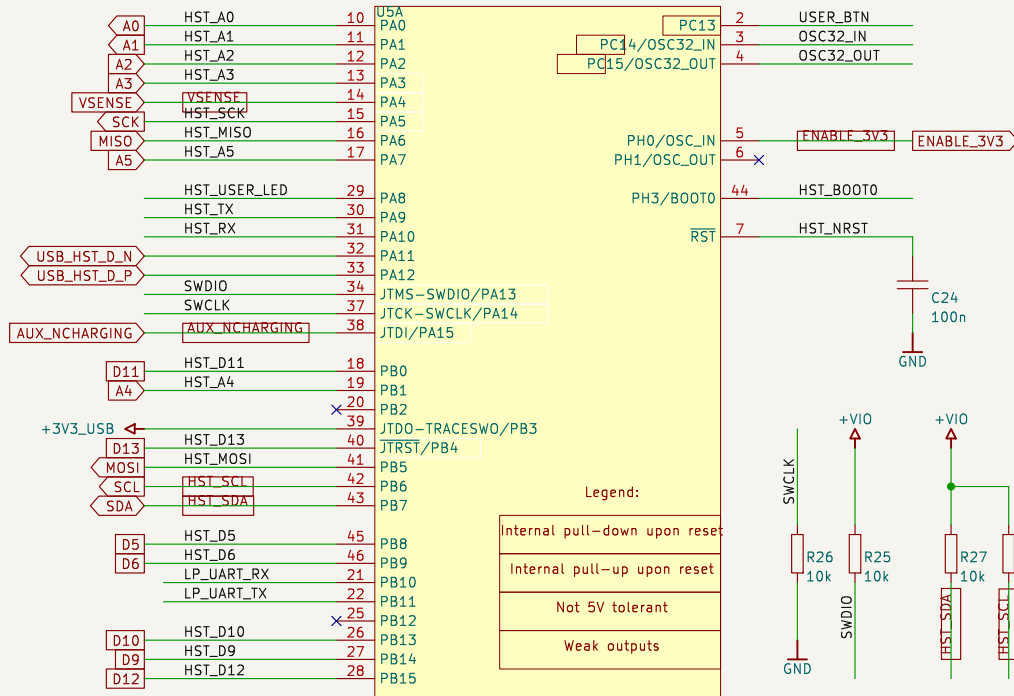
VDETECT



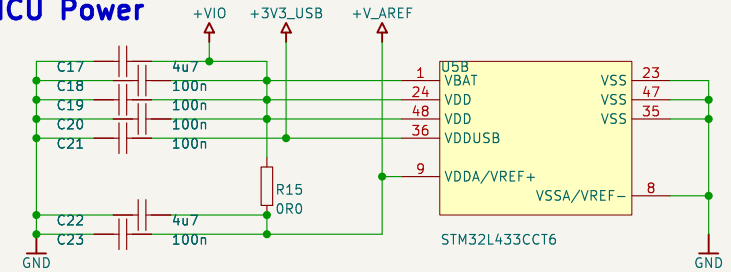
Title:	Notecarrier-CX.kicad_sch	Title:	
Project:	Notecarrier-CX	Drawn By:	
Company:	Medarska 69/1, 10000 Zagreb, Croatia	File:	Notecarrier-CX
Byte Lab Grupa d.o.o.	www.byte-lab.com	Rev:	HW_E.Doc_Rev
Author:	A. Duracic	Accepted:	E. Stanisha
Sheet 4 of 5		Rev:	HW_E.Doc_Rev



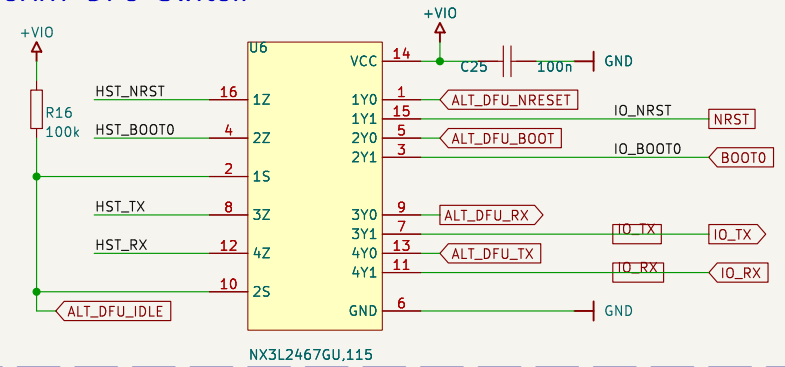
Host MCU



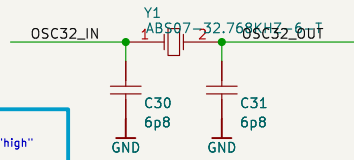
Host MCU Power



UART DFU Switch

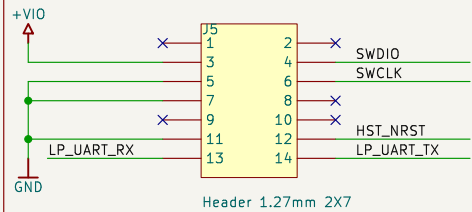


MCU Xtal

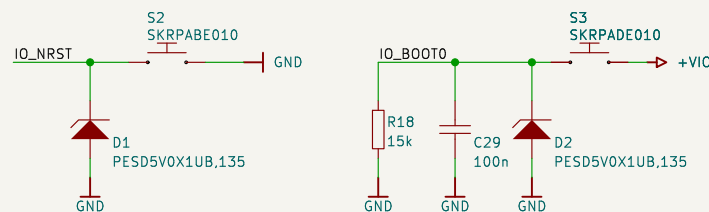


SOFTWARE CONFIG NOTE:
- Use drive level "medium high" or "high"
to ensure startup

Debug Connector



User Buttons



Title: Notecarrier-CX.kicad_sch	Title:
Project: Notecarrier-CX	Drawn By:
Company: Byte Lab Grupa d.o.o.	Medarska 69/1, 10000 Zagreb, Croatia
File: Notecarrier-CX	www.byte-lab.com
Author: A. Duracic	Rev: = HW_E.Doc, Rev: A4
Copyright (c) 2023 Blues Inc.	Accepted: E. Stanisa

